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An Improved A-Star Algorithm for Global Path Planning of Unmanned Surface Vehicle

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Abstract: Conventional A-Star (A*) algorithm exists some problems, such as multiple extended nodes, slow convergence and weak real-time performance, in marine path planning of unmanned surface vehicles (USVs). To solve the aforementioned problems, this study proposes an adaptive-direction and variable-step A* algorithm. Firstly, a gridded environment model is established based on S57 electronic navigation chart. Considering the marine scenes, the obstacles and water flow constraints are added into the evaluation function. Secondly, an adaptive direction method and a variable step strategy are proposed, and the bi-directional search method is combined with the proposed method to form an improved A* algorithm. An evaluation function for assessing redundancy is designed to optimize the path to suppress the influence of redundant and non-smooth paths. Finally, some visualized simulated experiments are set to verify the performance of the proposed method. And three different scenes ranging from simple to complex are selected for path planning experiments. The results show that the improved algorithm outperforms the conventional A* algorithm, artificial potential field algorithm and particle swarm optimization algorithm. It can effectively reduce extended nodes, and has good real-time performance. In addition, it has the robustness and faster convergence that can be used in different path-planning scenes.

Key Words: Global path planning, USV, Improved A* algorithm, Path optimization

1 Introduction

With the in-depth application of artificial intelligence technology in various fields, the research of intellectualized vehicles has become a hotspot[1]. Marine unmanned surface vehicle (USV) can autonomously complete scheduled tasks, and it is commonly applied in environmental monitoring, marine geodesy and cartography, search and rescue, accident emergency response and electronic counter--measures[2,3].

Global path planning ensures the executive efficiency of USV. It is to find an optimal path from the starting node to the target node based on the known map information[4]. Gong[5] combined the particle swarm optimization (PSO) algorithm and the differential evolution algorithm. This algorithm can adaptively adjust the mutation operator to maintain the global search ability and robustness of path planning. However, the swarm intelligence optimization algorithm is difficult to adjust parameters. It easily produces premature convergence to fall into local optimum. Furthermore, it takes a long time to run in large-scale maps, and may not generate feasible paths, when there are a lot of irregular obstacles. The artificial potential field (APF) algorithm[6] is to realize path planning based on the virtual gravitation and repulsive force fields, which are generated by the environment. For the shortcoming of the local minimum, Yang[7] modified the gravitational field and evaluation function, and solved the problem of unreachable targets by adding gravity. The experiments showed shorter running time and path lengths. However, the improved APF algorithm still has the problem that the virtual gravitation and repulsive force are difficult to balance. Dijkstra algorithm[8] is a typical single source shortest path algorithm based on the breadth-first search. However, the space-time complexity is high. On the basis of the Dijkstra algorithm, A-Star (A*) algorithm[9] introduces a heuristic function to improve search efficiency. Compared with other algorithms, it can achieve global optimization and has good stability. Therefore, this paper studies path planning based on A* algorithm. Liu[10] established risk models of water current, traffic separation and berthing for vehicle path planning. It realized a vehicle path planning, which approximates real scenes. However, the efficiency for the complex environment is still low. The conventional A* algorithm has many inflection nodes. To solve this problem, Ou[11] proposed a security protection mechanism and path smoothing strategy. And an adaptive cost function was considered to improve search efficiency. However, this algorithm extends a large number of nodes when the map size is large. Zhou[12] proposed a multiscale map method based on quad-tree, which effectively solved the problem of the A* algorithm wasting time in irrelevant areas. But this method requires a large amount of time and memory during initialization, and it may underperform in different application scenes.

In this study, an improved path planning method for USV, which is based on the A* algorithm, is proposed to solve lots of extended nodes, low efficiency and non-smooth path problems at large-scale scenes. The true information of the map is obtained from an electronic navigation chart. The obstacles constraint and the water flow influence are considered in the evaluation function. And an adaptive direction method and a variable step strategy are proposed to reduce the number of extended nodes. To increase the efficiency of the search, the proposed method is combined with the bi-directional search method to improve searching speed. Finally, an evaluation function for assessing

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redundancy is designed to reduce redundant path nodes to improve smoothness.

2 Environmental Model

The environmental model is a key step of global path planning. Compared with other maps, the electronic navigation chart has higher resolution and more marine information. To estimate reliable path planning results, and ensure high safety, this study chooses S57 electronic navigation chart to establish an environmental model.

The common environmental modeling methods include visibility graph[13], Voronoi diagram[14] and grid method[15]. The visibility graph method can search the globally optimal path. However, the complexity is in direct proportion to the number of obstacles, and the generated path may collide with obstacles. Voronoi diagram can guarantee the safety of the generated path. Due to the limitations of the search nodes, the generated path is long, and this method needs to rely on positioning sensors. The grid method is the most commonly used simple environmental modeling method. The high-resolution grid map can fully include information of obstacles, and the generated map is easy to tune to obtain different resolutions and sizes. Therefore, this method is selected to establish an environmental model. However, the search rate of the grid method is affected by grid density, and the generated sawtooth may cause a non-smooth path. These problems will be optimized by the proposed method.

The grid method divides continuous environment space into grids. In this study, the grid value is set to 0 or 1 to indicate the feasible and infeasible region. There are two methods to mark the gird position: coordinate and sequential methods. In $n \times n$ map, the corresponding relationship between the two ways[16] is shown in formula (1):

$$\begin{cases} x = \mathrm{mod}(N-1, n) + 0.5 \\ y = n + 0.5 - ceil(N/n) \end{cases} \tag{1}$$

where, (x, y) represents the coordinate, and N is the sequence number corresponding to the coordinate.

3 Improvement of A* Algorithm

Conventional A* algorithm has lots of extended nodes, low search efficiency and non-smooth path problems in the path planning research of USV in large-scale complex scenes. To solve these problems, an improved algorithm, which is called the adaptive-direction and variable-step A* (ADVSA*) algorithm, is proposed in this study. It proposes an adaptive search direction method and variable step strategy to reduce extended nodes and improve search efficiency, and designs an evaluation function for assessing redundancy to remove redundant nodes and smooth the path. At the same time, the bi-directional search method and an evaluation function considering the obstacles and water flow constraints are used in the improved A* algorithm.

3.1 Conventional A* Algorithm

The conventional A* algorithm inherits the Dijkstra algorithm, and it combines the best-first search and heuristic function. This characteristic maintains higher searching efficiency. The goal of the A* algorithm is to find a path with the minimum cost, by searching from the starting node to the target node. The flowchart of the conventional A* algorithm is denoted in Algorithm 1.

Algorithm 1 Conventional A* ($N_s \rightarrow N_t$)

```
Add N_s into openlist
while openlist ≠ ∅ do
    Find N_c with the minimum evaluation function value f(N_i) from
    the openlist
    if N_c = N_t then
        return "Path is found."
    else
        Add N_c into closelist
        for all neighbor nodes N_bi of N_c
            if N_bi is a feasible node and not in the closelist then
                Add N_bi into the openlist and calculate f(N_bi)
                if N_bi is in the openlist and g(N_bi) is smaller than g'(N_bi) of
                the previous N_bi then
                    f(N_i)=f(N_bi) and set the parent node of N_i as N_c
                end if
            end if do
        end for
    end if
end while
return "Path is not found."
```

The A* algorithm has two lists. The *openlist* records all the extended nodes of the searching process. The *closelist* records unconsidered nodes. The evaluation function $f(N_i)$ is calculated as formula (2):

$$f(N_i) = g(N_i) + h(N_i) \tag{2}$$

where $g(N_i)$ is the actual cost function of the path from starting node to the current node N_i, and $h(N_i)$ is the heuristic function of the path from N_i to the target node.

3.2 Bi-directional Search Method

When the map is large, the conventional A* algorithm extends many nodes, resulting in slow convergence. Therefore, the bi-directional search method is introduced to improve search efficiency.

The bi-directional search method uses two A*-algorithm searching strategies, where the forward A* algorithm searches from the starting node and the reverse A* algorithm searches from the target node. The search ends when there are repeated nodes in the *closelist* of two A* algorithms. And a feasible search path can be obtained by combining nodes in the *closelist* of two A* algorithms. Compared with the conventional A* algorithm, searching from the starting node and the target node can improve the search efficiency and find a search path faster.

3.3 Improvement of Evaluation Function

The search of A* algorithm is based on the evaluation function, and an appropriate evaluating criterion is important to find the optimal path. This study improved the evaluation function by considering the obstacles constraint and the water flow influence. The evaluation function is defined as formula (3):

$$f(N_i) = g(N_i) + h(N_i) - c(N_i) \tag{3}$$

Formula (4) is the heuristic function $h(N_i)$, which consists of obstacles constraint penalty and distance value.

$$h(N_i) = obs(N_i) \times Diag(N_i) \tag{4}$$

Formula (5) is obstacles constraint penalty $obs(N_i)$:

$$obs(N_i) = P \times \sum_{j=1}^{n} sign(D - \sqrt{(x_i - ox_j)^2 + (y_i - oy_j)^2}) \tag{5}$$

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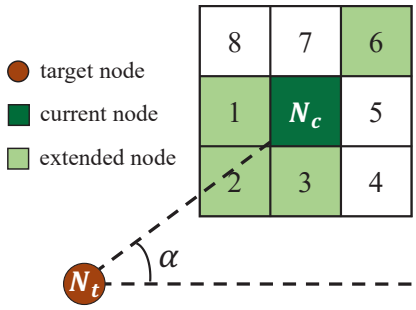

Fig. 1: Adaptive search direction selection method

Table 1: The Rule of The Direction Selection

dir	Node number
D_1	1, 2, 3, 6
D_2	3, 4, 5, 8
D_3	5, 6, 7, 2
D_4	7, 8, 1, 4

where P is the penalty coefficient. D is the safety distance value. When the distance between the current node and the obstacle node is less than D, there is a non-zero obstacle penalty value. (x_i, y_i) is the current node, (ox_j, oy_j) (j = 1, 2, …, n) is the obstacle node.

The method of calculating the distance of the heuristic function includes Manhattan distance, Euclidean distance and Diagonal distance. Considering the balance between extended direction and running time, this study selects the Diagonal distance method. The Diagonal distance[17] of node N_i is as formula (6):

$$Diag(N_i) = 1.4 \times \min(|x - x_i|, |y - y_i|) + ((|x - x_i| + |y - y_i|) - 2 \times \min(|x - x_i| + |y - y_i|)) \tag{6}$$

where, (x_i, y_i) is the current node. When $h(N_i)$ is the heuristic function of the forward A* algorithm, (x, y) is the current node of the reverse A* algorithm. When $h(N_i)$ is the heuristic function of the reverse A* algorithm, (x, y) is the current node of the forward A* algorithm.

Formula (7) is the water flow influence:

$$c(N_i) = v_{ie} \times \cos(\theta_{id} - \theta_{ie}) \tag{7}$$

The calculation is based on the connection between the current node N_i and water flow. v_{ie} is the water flow velocity of N_i. θ_{id} is the angle between N_i and the x axis, and θ_{ie} is the angle between water flow direction of N_i and the x axis.

3.4 Adaptive Direction Method

The conventional A* algorithm usually uses 4 or 8 neighborhood searches, which leads to a large number of extended nodes and long computing time. According to the position relationship between the current node and the target node, this study proposes an adaptive direction method. The method can extend nodes more purposefully to reduce search nodes and accelerate the convergence speed.

As shown in Fig. 1, taking the target node N_t as the center, and the angle between the current node N_c and N_t is calculated. Then select three neighbor nodes N_i (i = 1, 2, 3) of N_c, which are closest to N_t as extended nodes. However, when there are large obstacles between the starting node and the target node, the algorithm search may fall into the dead zone. Therefore, a reverse node is added into N_i to avoid this problem. Finally, calculate the evaluation function $f(N_i)$ of neighbor nodes, and determine the extended direction in the next step.

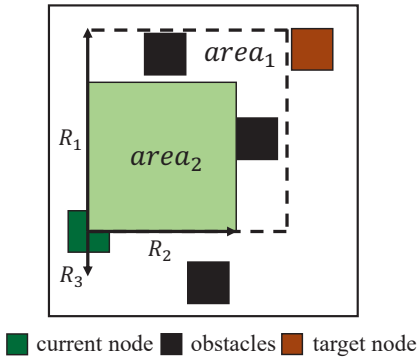

Fig. 2: Variable step selection strategy

Formula (8) is the angle between the target node N_t and the current node N_c. (N_{tx}, N_{ty}) is the coordinate of the target node, and (N_{cx}, N_{cy}) is the coordinate of the current node.

$$\alpha = \arctan \frac{N_{ty} - N_{cy}}{N_{tx} - N_{cx}} \tag{8}$$

The corresponding relationship between the direction selection and the angle is as formula (9):

$$dir = \begin{cases} D_1, & 0^\circ \le \alpha < 90^\circ \\ D_2, & 90^\circ \le \alpha < 180^\circ \\ D_3, & 180^\circ \le \alpha < 270^\circ \\ D_4, & 270^\circ \le \alpha < 360^\circ \end{cases} \tag{9}$$

Table 1 shows the specific rule of direction selection, corresponding to neighbor node numbers in Fig. 1.

3.5 Variable Step Strategy

The conventional A* algorithm extends continuous nodes during the search process. When the map has a large feasible region, many redundant nodes will be extended, resulting in long search time. Therefore, this study proposes a variable step strategy. The search step value is determined based on the surroundings of the current node N_c, thereby reducing the number of extended nodes and improving search efficiency.

As shown in Fig. 2, within the range of $area_1$, if there are no obstacles or target node N_t, the step value is R_1. Otherwise, the environment within $area_2$ is judged. If there are no obstacles or the target node, the step value is R_2. Otherwise, the step value is R_3, which means the obstacles in the current environment are dense.

Formula (10) is the calculation method of step value.

$$step = \begin{cases} R_1, & area_1 \cap obs = \varnothing \wedge N_t \notin area_1 \\ R_2, & area_2 \cap obs = \varnothing \wedge N_t \notin area_2 \\ R_3, & otherwise \end{cases} \tag{10}$$

where, R_1, R_2 and R_3 is the step value, and obs are all obstacles. N_t represents the target node. $area_1$ and $area_2$ are regions that need to be judged whether it is feasible or not. Defined as formula (11):

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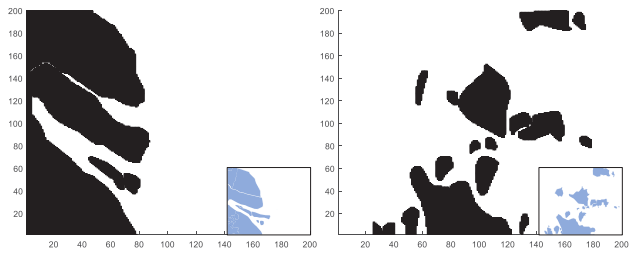
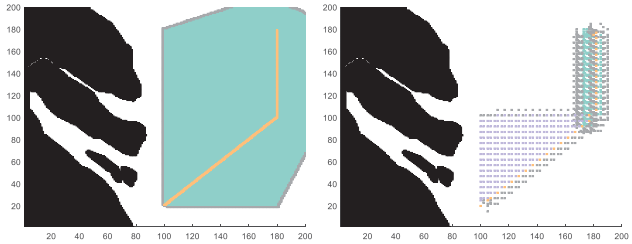


(a) map_1 (b) map_2

Fig. 3: Environment model



(a) cA* algorithm (b) ADVSA* algorithm

Fig. 4: Path search process in $Path_1$

$$area_i = MAP(N_{cx}, N_{cy}, N_{cx} + R_i \times \mathrm{sgn}(\cos(\alpha)), \\ N_{cy} + R_i \times \mathrm{sgn}(\sin(\alpha))), \quad (i = 1, 2) \tag{11}$$

where, MAP is the map data used in path planning, and (N_{cx}, N_{cy}) is the current node coordinates. α is the angle between the target node and the current node, which can be calculated by formula (8).

3.6 Path Optimization

Due to the A* algorithm based on the grid method for path planning, the path results will exist sawtooth and redundant nodes. To solve these problems, an evaluation function for assessing redundancy is proposed to optimize the path, which can remove redundant path nodes, eliminate sawtooth, and improve the smoothness of the path.

The steps of path optimization are as follows:

*step*1: Based on the path planning result, find the turning nodes set $Z = \{z_1, z_2, \ldots, z_l\}$;

*step*2: Computer the evaluation function for assessing redundancy of each element in Z. The formula (12) is the evaluation function for assessing redundancy. z_l represents the current turning node, and (dx_i, dy_i) is m nodes obtained by linear interpolation from z_{l-1} node to z_{l+1} node. (ox_j, oy_j) (j = 1, 2, …, n) is the obstacle node.

$$S(z_{l-1}, z_{l+1}) = \sum_{i=1}^{m}\sum_{j=1}^{n}\sqrt{(dx_i - ox_j)^2 + (dy_i - oy_j)^2} \tag{12}$$

*step*3: Judge whether the turning nodes are redundant nodes according to calculation results. If $S = 0$, it means that z_{l-1} node is connected to z_{l+1} node without obstacles, and z_l node is a redundant node that can be removed. Otherwise, z_l node needs to be retained. The turning nodes set Z with redundant nodes removed is the result of path optimization.

4 Simulation Experiment

4.1 Environmental Model

Based on Visual Studio 2019, the electronic navigation chart parsing algorithm is used to analyze the chart file of C1100103. Select two areas of the East China Sea from the analytic file to build gridded environment models. The range of map_1 is (31.11°N~32.11°N, 121.17°E~122.96°E), and set the resolution is (0.005°, 0.009°) to generate a map of 200×200, as shown in Fig. 3 (a). The range of map_2 is (30.4474°N~30.07°N, 121.7°E~122.6°E), and set the resolution is (0.0017°, 0.0045°) to generate a map of 200×200, as shown in Fig. 3 (b). The blue part is the real map of the region.

4.2 Experimental Setup

In order to illustrate the performance of the proposed algorithm, three representative paths are selected for path planning experiments. $Path_1$ is based on map_1, which represents a simple scene. The grid coordinates of the starting node and target node are (180, 180) and (100, 20), respectively. $Path_2$ is based on map_1, which represents a complex scene with obstacles at both ends. The grid coordinates of the starting node and target node are (70, 190) and (70, 30), respectively. $Path_3$ is based on map_2, which represents a complex scene with obstacles in the middle. The grid coordinates of the starting node and target node are (6, 120) and (180, 60), respectively.

The simulation experiments are carried out on MATLAB R2021a. To simplify description, the conventional A* algorithm is donated as cA*. The parameters of the ADVSA* algorithm are as follows: the penalty coefficient P is set to 100, and the safety distance value D is 1. The water flow velocity v_{ie} is 1.4. The step values R_1, R_2 and R_3 are 5, 2 and 1, respectively.

4.3 Simulation Result

To illustrate the effectiveness of the improved method, the path search processes of cA* and ADVSA* algorithm in $Path_1$ are compared, as shown in Fig. 4. Blue nodes are the points that algorithm extends from the starting node, and purple nodes are the points that algorithm extends from the target node. Gray nodes are the points that algorithm will extend in the next step, and orange nodes are the final path nodes obtained by searching. It can be seen that the cA* algorithm extends a lot of nodes in path planning, and the nodes are continuous, resulting in low search efficiency. By adopting the adaptive direction method and variable step strategy, the ADVSA* algorithm greatly reduces the extension of redundant nodes. At the same time, the bi-directional search method extends from the starting node and the target node, thus improving the search efficiency.

After 20 repeated experiments, Fig. 5 and Table 2 show the results of the ADVSA* algorithm before and after path optimization. The influence of water flow on path planning is considered in experiments. The blue line is the path before optimization, and the red line is the path after optimization. Through analyzing the results, path optimization in all scenes can effectively improve path smoothness and shorten path length, which indicates the effectiveness of path optimization.

In addition, the ADVSA* algorithm is compared with the cA*, APF and PSO algorithms, to demonstrate the good performance of the algorithm for global path planning. Con-

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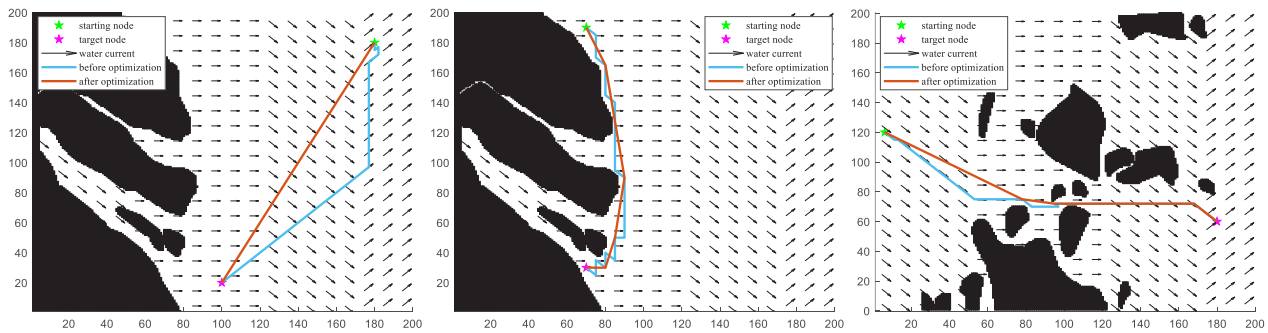

Fig. 5: ADVSA* algorithm before and after path optimization (*Path*$_1$, *Path*$_2$, and *Path*$_3$, respectively)

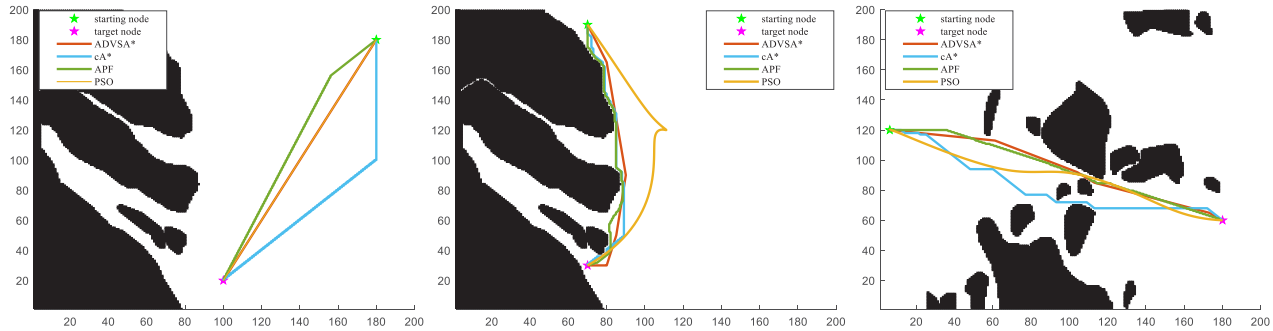

Fig. 6: Comparison of four path planning algorithms (*Path*$_1$, *Path*$_2$, and *Path*$_3$, respectively)

Table 2: Path Length of ADVSA* Algorithm Before and After Path Optimization

Path num	Before	After
Path$_1$	198.79	**178.89**
Path$_2$	209.5	**173.52**
Path$_3$	204.51	**192.14**

-duct 20 repeated experiments, and the results of running time and path length are shown in Fig. 6 and Table 3.

It can be seen that in *Path*$_1$ and *Path*$_3$, the running time and path length of the improved algorithm are both optimal, indicating that the improved algorithm has good convergence. Although cA* and APF algorithms are short in running time, the results of path length are worse than the ADVSA* algorithm. And the smoothness of cA* and APF is poor. Limited by the turning restriction of the USV, they cannot be well applied to practice. Although the path length of the PSO algorithm in *Path*$_1$ is the same as the ADVSA* algorithm, it does not perform well in *Path*$_3$. And the running time is non-ideal. This is because it is easy to fall into the local optimal, resulting in a long search time and poor convergence. Due to the introduction of the path optimization method, the computation complexity of the improved algorithm increases. Therefore, in a complex scene with dense obstacles such as *Path*$_2$, the running time of the ADVSA* algorithm is slightly longer than the cA* algorithm, but the path length is still optimal.

The experiments show that the ADVSA* algorithm has good convergence, and can meet real-time requirements. Meanwhile, the robustness and applicability are verified, which can be applied to path planning in various scenes.

5 Conclusion

In this study, an adaptive-direction and variable-step A* (ADVSA*) algorithm is proposed for marine global path planning of USVs. Firstly, a gridded environment model is established based on the S57 electronic navigation chart. Considering the marine scenes, the obstacles and water flow constraints are added into the evaluation function. Then an adaptive direction method and variable step strategy are proposed, and the bi-directional search method is combined with the proposed method to form the improved algorithm. Finally, this study designs an evaluation function for assessing redundancy to optimize the path. The path planning experiments selected three different scenes from simple to complex. The search processes of algorithms are visualized, to indicate the improved algorithm can greatly reduce the extended nodes and improve search efficiency. To illustrate the effectiveness of the path optimization method, the ADVSA* algorithm before and after path optimization is compared. Then the ADVSA* algorithm is compared with the conventional A* algorithm, artificial

Table 3: Results of Four Path Planning Algorithms

Path num	Algorithm	Running time	Path length
Path$_1$	ADVSA*	**0.14**	**178.89**
	cA*	1.48	240.00
	APF	1.90	192.00
	PSO	32.89	178.89
Path$_2$	ADVSA*	0.29	**173.52**
	cA*	**0.27**	186.87
	APF	1.77	177.00
	PSO	32.12	186.40
Path$_3$	ADVSA*	**0.14**	**186.55**
	cA*	0.22	198.85
	APF	0.99	197.00
	PSO	16.73	203.06

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potential field method and particle swarm optimization algorithm. The results show that the improved algorithm can effectively reduce the expansion nodes, and has good real-time performance. In addition, it has the robustness and faster convergence that can be used in different path-planning scenes.

In the future, we will consider including vehicle characteristics or various environmental disturbances in the evaluation function, so as to study marine path planning that is closer to the real environment.

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